Causal Inference for Panel Data

An applied approach
Day 1

Dr. Valentina Di Iasio

9th and 10th December 2025

University of Southampton

Course introduction

Who I am

- Research Fellow in Economics at the University of Southampton
- Applied Economist with research interest in Migration and Development

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    - Main issues and problems often encountered in applied work
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- The main aim of this course is to share some of my experience with you:
    - Main issues and problems often encountered in applied work
    - Some tips to overcome them
- We will treat all the topics of this course mainly from an **applied perspective**

Overview of the course

This course is organised over 2 days:

Day 1

- December 9th from 9:45 to 12:15(ish)
- Building 58 Room 2097 and online
- **Topic 1:** Definition and features of panel data
- **Topic 2:** Difference-in-differences: introduction and review of the up-to-date literature
- **Topic 3:** Longitudinal datasets available on the UK (optional)

Overview of the course

Day 2

- December 10th from 10 to 13
- Building 58 Room 2097 and online
- Lab session using **Stata**
- **Part 1:** Problem set 1 - guided
- **Part 2:** Problem set 2 - guided
- **Part 3:** Problem set 3 - autonomous

Today's schedule

Day 1 - Schedule

- 9:45 $\rightarrow$ Introduction to this course
- 10:00 $\rightarrow$ Panel data - why so useful for causal inference
- 10:45 $\rightarrow$ DiD - definition and assumptions
- 11:15 $\rightarrow$ Break
- 11:30 $\rightarrow$ DiD in practice - overview of the new literature
- 11:45 $\rightarrow$ UK longitudinal datasets (optional)

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Otherwise it becomes a boring monologue :)

Table of contents

The issue of causality

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- How public policies affect poverty/unemployment etc?
- Does higher education lead to better jobs?
- Will attend this Winter School make me a more successful researcher?

The issue of causality

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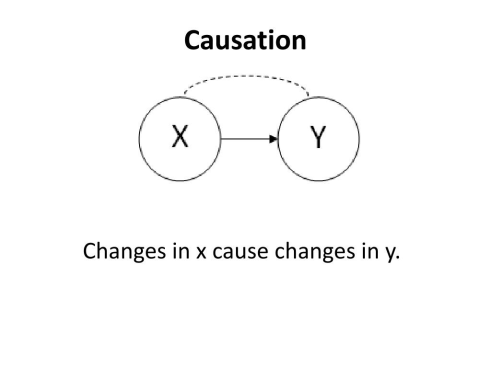
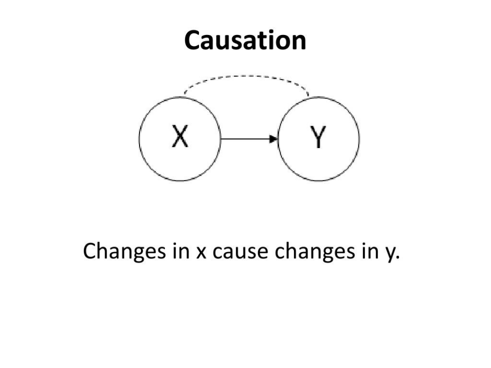


The issue of causality

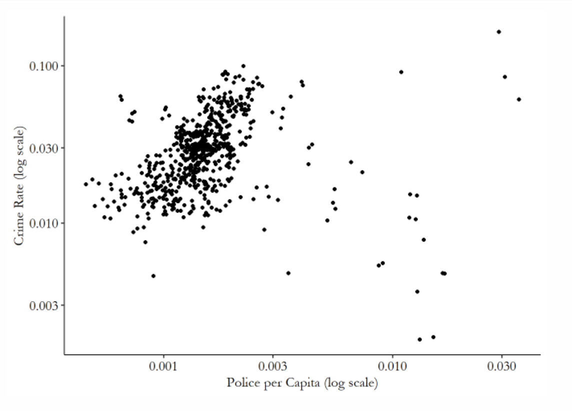


E.g. Sending additional police in the streets will decrease crime

The issue of causality

However, reality is much more complex:

Figure 1: Police Presence and Crime Rate by County, North Carolina 1981-1987



Source: theeffectbook.net

Panel data - definition

We need data

Experimental data (RCT)

- Ideal scenario
- Designed to evaluate a specific policy/treatment and investigate causal effect

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- Ideal scenario
- Designed to evaluate a specific policy/treatment and investigate causal effect
- Often not feasible and difficult to set up
- Can present issues of external validity

We need data

Observational data

- Data obtained outside an experimental setting
- Survey data, administrative data, historical records
- Combine different econometrics techniques with clear identification strategy
- Mostly used in applied work

We need data

Quasi-Experimental Approaches

- Use observational data and find ”natural experiments”
- Need to motivate identification strategy and find a clear exogenous variation
- Useful econometrics tool: Difference-in-differences, RDD, IV

Cross-sectional data

- Multiple units, only one time

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Notation

Y_i and X_i where $i = 1, \ldots, N$

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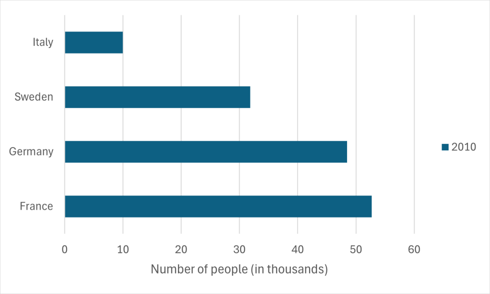
Figure 2: Total number of asylum applicants in 2010 in selected EU countries



Source: Asylum applicants by type, citizenship, age and sex - annual aggregated data, Eurostat

Time series data

- Single unit, multiple times

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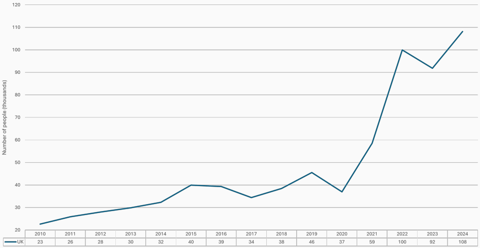
Figure 3: Trends in total number of people claiming asylum in the UK



Source: Asy_D01 - Asylum applications raised, Home Office

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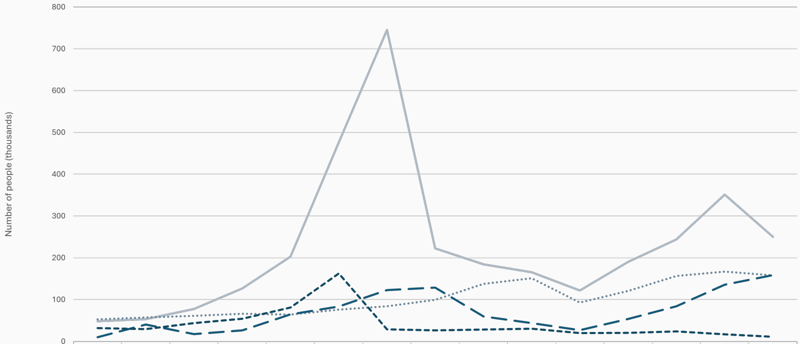
Panel data - example

Figure 4: Trends in total number of asylum applicants in selected EU countries

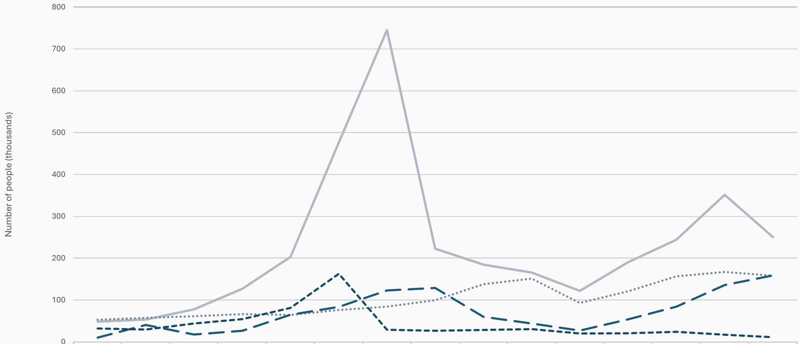


	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Germany	48.48	53.24	77.485	126.705	202.645	476.51	745.16	222.565	184.18	165.615	121.955	190.545	243.835	351.51	250.55
France	52.725	57.33	61.44	66.265	64.31	76.165	84.27	99.33	137.665	151.07	93.2	120.685	156.455	166.94	157.85
Italy	10	40.315	17.335	26.62	64.625	83.54	122.96	128.85	59.95	43.77	26.94	53.61	84.29	135.82	158.605
Sweden	31.85	29.65	43.855	54.27	81.185	162.45	28.795	26.33	28.605	30.325	20.195	20.375	23.805	17.01	11.11

Source: Asylum applicants by type, citizenship, age and sex - annual aggregated data, Eurostat

Panel data - balanced

Figure 5: Trends in total number of asylum applicants in selected EU countries



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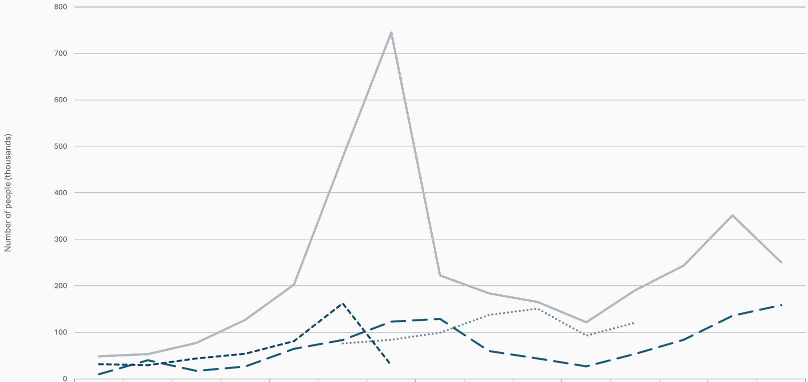
Panel data - unbalanced

Figure 6: Trends in total number of asylum applicants in selected EU countries



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Definitions

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A panel is balanced if we have the same time periods, $t = 1, \ldots, T$, for each cross section observation. A panel is unbalanced if the time dimension is not the same to each individual.

- Issue of sample selection and attrition
- Attrition: when individuals drop out of the sample

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Micro vs Macro

- Micro: More i than t
- Macro: More t than i or roughly the same

Panel data in practice

Panel data: main advantage

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Main model

$$Y_{it} = \alpha_i + \beta_1 X_{it} + \epsilon_{it} \tag{1}$$

OLS is only consistent if X_{it} is uncorrelated with the error ϵ_{it}

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Fixed effects will only eliminate μ_i, the part of the error that is constant over time

Defining your equation in practice

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- Use one of the software commands (recommended)

In practice using Stata

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- Control for the unit-invariant heterogeneity related to the time unit
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- With xtreg need to include time dummies (e.g. i.year)

TWFE equation

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xtreg *depvar indepvar*, fe vce(cluster id)

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- More computationally efficient
- Stata command $\rightarrow$ reghdfe *depvar indvar, absorb() vce()*

Other advantages

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- Virtually more observations
- Multiple points in time $\rightarrow$ you can use leads and lags
- You can consider using random effect to estimate causal relationships if you have multi-level data $\rightarrow$ more on this

Caveats

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- Fixed effects would only estimate a causal effect if the omitted variable bias only stems from time-invariant unobservable heterogeneity
- With non-linear model $\rightarrow$ incidental parameter problem

Practical example

I want to estimate the causal effect of the number of published papers on the probability of getting a job in academia

$$Y_{it} = \alpha_i + \beta_1 X_{it} + \epsilon_{it} \tag{6}$$

- Which variables would you include as control variables?
- What could be an example of time-invariant heterogeneity that can be difficult to observe in real data?
- And time variant?
- Are there any issues of reverse causality in this relationship?

Difference-in-differences

DiD - Introduction

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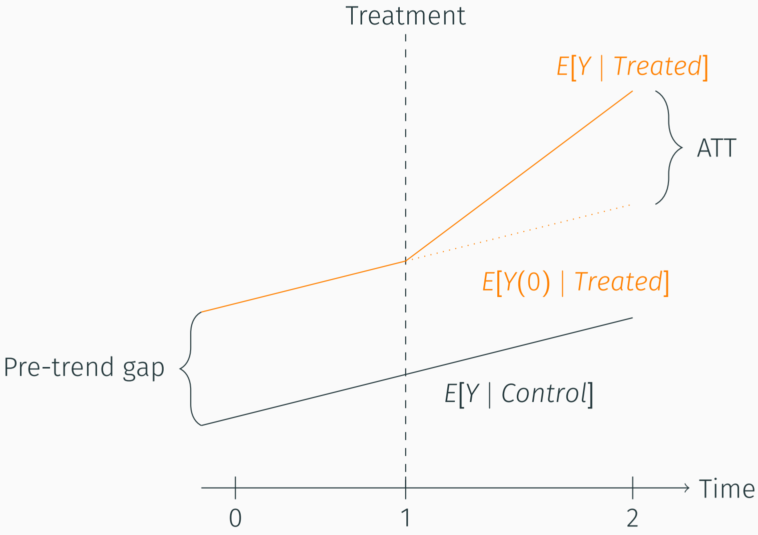
- Observe the outcomes of treatment and control group before and after the policy/intervention took place
- Need panel data

Short video

DiD introduction by Angrist

DiD what are we estimating?



Source: Adapted from **Mixtape - Advanced DiD**

Standard DiD in practice

$$Y_{it} = \alpha + \beta_1 Treated_i + \beta_2 Post_t + \beta_3 (Treated_i \times Post_t) + \epsilon_{it} \qquad (7)$$

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- The DiD is the gap in the slopes

Standard DiD - TWFE

$$Y_{it} = \beta_3(Treated_i \times Post_t) + \mu_i + \delta_t + \epsilon_{it} \tag{8}$$

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- δ_t = time fixed effects $\rightarrow$ controls for shocks common to all units at time t
- β_3 = Coefficient of the interaction $\rightarrow$ DiD effect

Elements of the standard DiD model

- No anticipation

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- All units treated at the same time

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- Absence of pre-trends $\rightarrow$ key identifying assumption
    - Visual inspection of pre-treatment trends
    - Event-study models
    - Placebo tests

Common ways to assess parallel trends

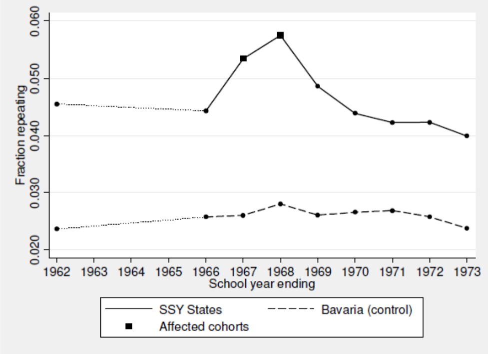
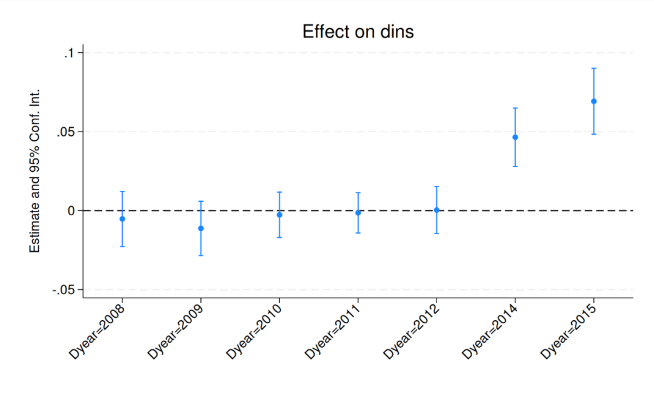


Figure 5.2.3: Average rates of grade repetition in second grade for treatment and control schools in Germany (from Pischke 2007). The data span a period before and after a change in term length for students outside of Bavaria.

Source: Mostly Harmless Econometrics

Common ways to assess parallel trends



Source: Honest DiD - Vignettes

Common ways to assess parallel trends

Placebo tests is essentially a *fake treatment*

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Common ways to assess parallel trends

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- Pick either: a period before the actual treatment; a variable different from the treatment; an outcome that shouldn't be affected by the treatment
- Run the DiD as if it was the real treatment
- If you find a non-significant (near zero) estimate: it supports the credibility of your estimations
- If you find a significant effect: it suggests that your DiD may be picking up pre-existing trends rather than the effect of your treatment

What if I have pre-trends?

Table 1
A checklist for DiD practitioners.

- Is everyone treated at the same time?

If yes, and panel is balanced, estimation with TWFE specifications such as (5) or (7) yield easily interpretable estimates.

If no, consider using a "heterogeneity-robust" estimator for staggered treatment timing as described in Section 3. The appropriate estimator will depend on whether treatment turns on/off and which parallel trends assumption you're willing to impose. Use TWFE only if you're willing to restrict treatment effect heterogeneity.

- Are you sure about the validity of the parallel trends assumption?

If yes, explain why, including a justification for your choice of functional form. If the justification is (quasi-)random treatment timing, consider using a more efficient estimator as discussed in Section 6.

If no, consider the following steps:

1. If parallel trends would be more plausible conditional on covariates, consider a method that conditions on covariates, as described in Section 4.2.
2. Assess the plausibility of the parallel trends assumption by constructing an event-study plot. If there is a common treatment date and you're using an unconditional parallel trends assumption, plot the coefficients from a specification like (16). If not, then see Section 4.3 for recommendations on event-plot construction.
3. Accompany the event-study plot with diagnostics of the power of the pre-test against relevant alternatives and/or non-inferiority tests, as described in Section 4.4.1.
4. Report formal sensitivity analyses that describe the robustness of the conclusions to potential violations of parallel trends, as described in Section 4.5.

- Do you have a large number of treated and untreated clusters sampled from a super-population?

If yes, then use cluster-robust methods at the cluster level. A good rule of thumb is to cluster at the level at which treatment is independently assigned (e.g. at the state level when policy is determined at the state level); see Section 5.2.

If you have a small number of treated clusters, consider using one of the alternative inference methods described in Section 5.1.

If you can't imagine the super-population, consider a design-based justification for inference instead, as discussed in Section 5.2.

Source: Roth et al. (2023)

Instructions for tomorrow

Before the session

- Create an account in UK Data Service: **Instructions**
- How to download data: **Tutorial**
- Study number: 8715
- Download the Stata version

UK longitudinal data

Aim of this session

Working as a Research Fellow I had the experience to work with different UK datasets (e.g. Understanding Society, UK Labour Force Survey, Next Steps, British Social Attitudes Survey)

Share my experience and a few tips $\rightarrow$ all that I would have liked to know before I started

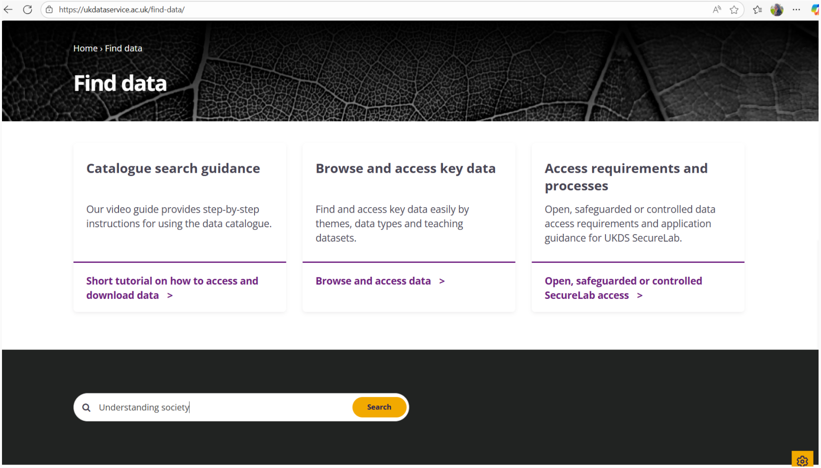
Hopefully to help you save some time and hassle

Survey data

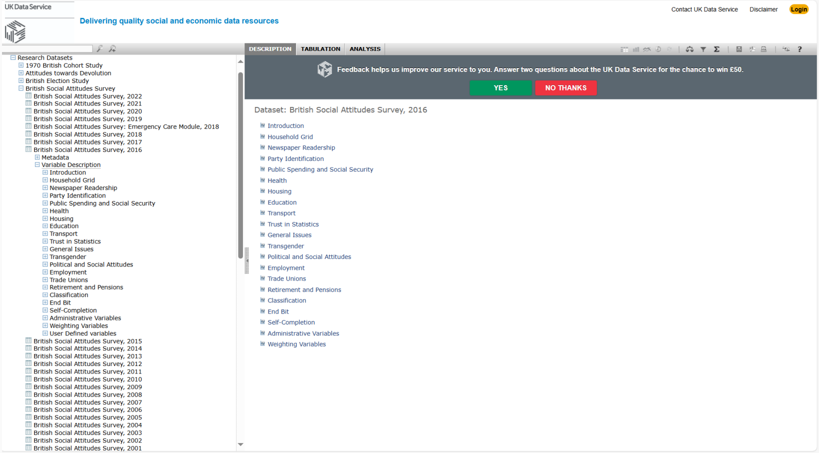
- Very good survey data available on the UK
- Examples of main longitudinal data $\rightarrow$ BHPS and Understanding Society, UK LFS[1]
- Longitudinal cohort studies $\rightarrow$ e.g. Next Steps, Millennium Cohort Study
- How to access them: the UK Data Service is one of the main data repository and a great source of information and training!

[1] Although this is a rotating panel

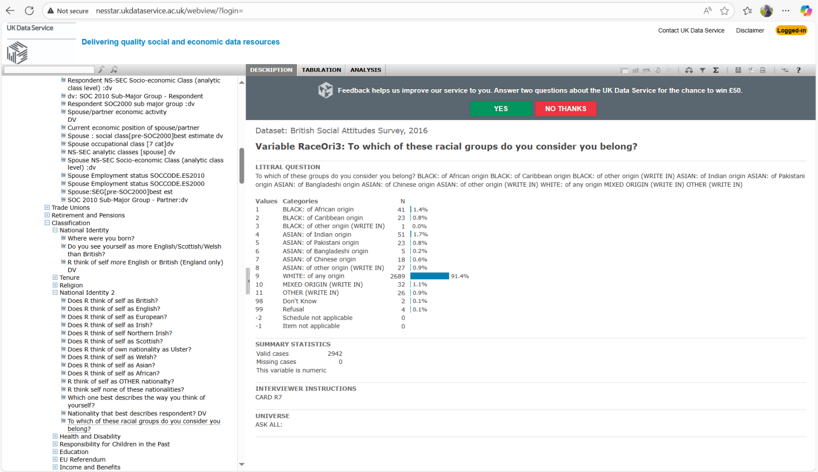
UK Data Service - find your data



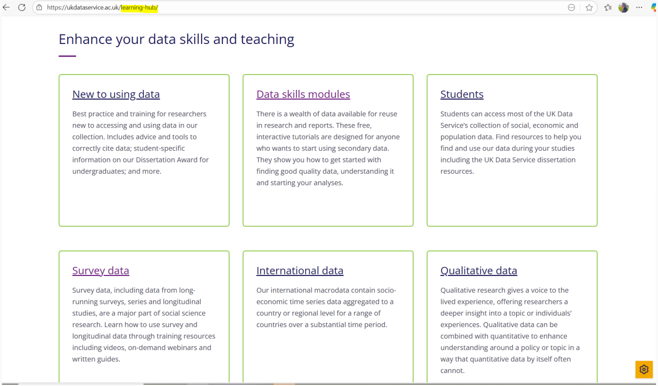
Nesstar - explore variables



Nesstar - explore sample



UK Data Service - training



UK Data Service - training

https://ukdataservice.ac.uk/training-events/

OCT 1 2025, 12:00 PM

Survey data drop-in

ONLINE

DROP-IN SESSION SURVEYS OPEN DOOR

OCT 2 2025, 10:00 AM

Introduction to effective and practical research data management

ONLINE

TRAINING RESEARCH DATA MANAGEMENT WORKSHOP

OCT 6 2025, 10:00 AM

Safe Researcher Training

ONLINE

TRAINING SAFE RESEARCHER WORKSHOP

OCT 7 2025, 1:30 PM

Code Anxiety Club

ONLINE

DROP-IN SESSION DATA SKILLS OPEN DOOR

About the safe researcher training...

Not all data are free to access, and in certain circumstances you will need to file an application and/or be an accredited researcher. The level of access depends on the sensitivity/confidentiality of the data.

- Open data $\rightarrow$ no registration needed
- Safeguarded data $\rightarrow$ only need to sign the End Users Licence Agreement
- Controlled data: within these we have
  - Secure Licence Data $\rightarrow$ need to file an application and sign the Secure Licence Agreement
  - Secure Access data $\rightarrow$ need to file an application and to complete the Safe Researcher Training (SRT).

More on the Secure Access data 1/2

- Applications for Secure Access data usually take time $\rightarrow$ make sure to factor it in your research plan
- All members of your research team have to complete the SRT (even if they do not plan to access the data)
- Once you are application is successful, you'll only be able to access the data using the Safe Research Environment.

More on the Secure Access data 2/2

Depending on the data you need, you'll either:

- Access the data using a University device which has been checked and approved during the application (needs a static VPN)
- Access the data using a Safe Pod/Safe Point

In any case the output of the analysis must be cleared before it can be shared. This also applies to your research team, i.e. if your research team does not access the data, you can't share the analysis output with them before clearance.

Some tips on the application process

The application process is generally time consuming. Some advice that could make it a bit smoother:

- Reply to all application's questions and items and make sure to download the latest application version
- Be very specific in your application (e.g. state your research questions and hypothesis very clearly)
- Specify the study number and name rather than just the study name (e.g. SN 8656 Next Steps: Sweeps 1-9, 2004-2023: Secure Access rather than Next Steps Secure Access)
- Apply for ethical approval at your University beforehand
- Make the most of your Safe Research Training
- Be prepared to revise it

Some general tips

- Make the most of your Safe Research Training also in preparing the output to be cleared if working with Secure data
- When working with survey data, make sure to use the appropriate weights in your analysis
- Data specific websites are a very useful source of information (e.g. Understanding Society website or UCL Center for Longitudinal Studies)
- Make sure you are familiar with the survey guideline and documentation

Administrative data

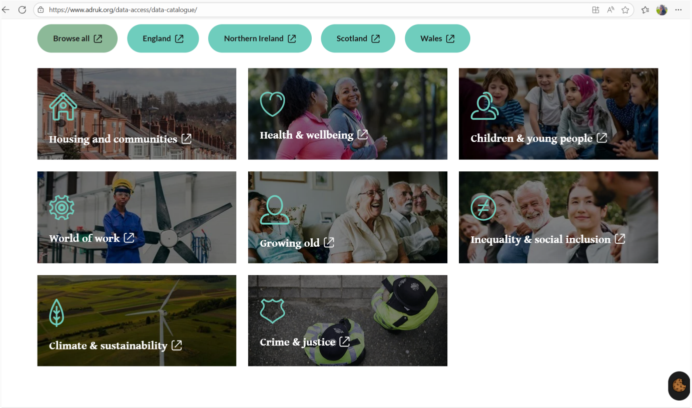
Administrative data is information created when people interact with public services, such as schools, the NHS, the courts or the benefits system, and collated by government.

These public bodies must keep records of these interactions for operational purposes: to enable them to carry out their day-to-day work, to monitor and improve their performance, and keep providing services in an effective way.

Pro and cons

- Important source of information for social sciences
- Some examples: Longitudinal Education Outcomes (LEO data), Annual Survey of Hours and Earnings (ASHE) linked to HMRC and 2011 Census
- Not self-reported $\rightarrow$ less measurement error
- Bigger numbers $\rightarrow$ more precise estimations and more possibility to conduct sub-group analysis
- More confidentiality issues $\rightarrow$ controlled data

Administrative Data Research UK

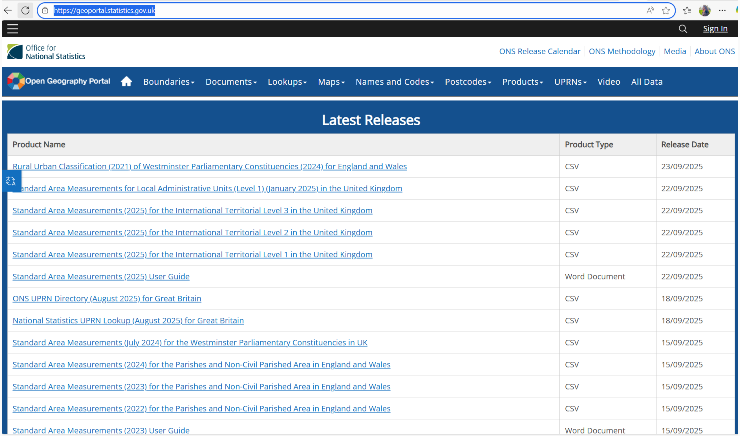


More on admnistrative data

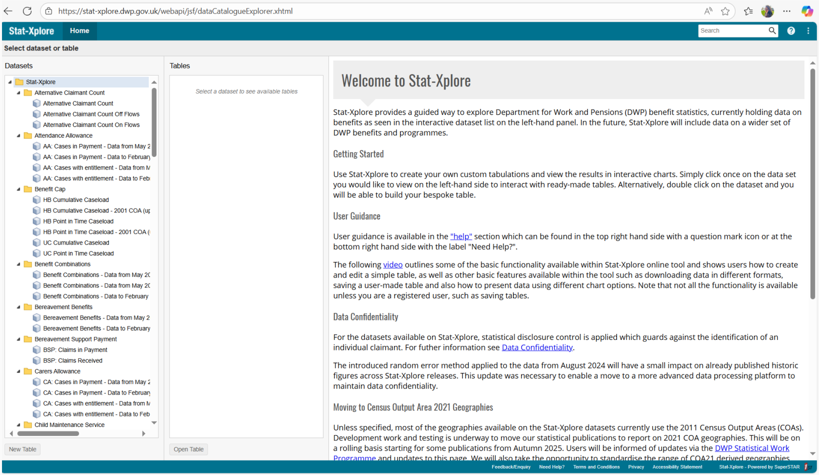
- Mainly available through Safe Pods or trusted research environment
- Data provider is the ONS
- Application process similar to the survey data Secure Access, slightly more complicated $\rightarrow$ more time consuming
- Apparently quite complicated to process and analyse
- Only available to Accredited Researchers

Geography data

- Often survey and/or administrative data will allow us to carry out analysis at a lower geographical level (e.g. regions or Local Authority Districts)
- Main source:



StatXplore


https://stat-xplore.dwp.gov.uk/webapi/jsf/dataCatalogueExplorer.xhtml
Stat-Xplore
Home
Search
Select dataset or table
Datasets
Stat-Xplore
Alternative Claimant Count
Alternative Claimant Count
Alternative Claimant Count Off Flows
Alternative Claimant Count On Flows
Attendance Allowance
Benefit Cap
HB Cumulative Caseload
HB Point in Time Caseload
UC Cumulative Caseload
UC Point in Time Caseload
Benefit Combinations
Bereavement Benefits
Bereavement Support Payment
BSP: Claims in Payment
BSP: Claims Received
Carers Allowance
Child Maintenance Service
New Table
Tables
Select a dataset to see available tables
Open Table
Welcome to Stat-Xplore
Stat-Xplore provides a guided way to explore Department for Work and Pensions (DWP) benefit statistics, currently holding data on benefits as seen in the interactive dataset list on the left-hand panel. In the future, Stat-Xplore will include data on a wider set of DWP benefits and programmes.
Getting Started
Use Stat-Xplore to create your own custom tabulations and view the results in interactive charts. Simply click once on the data set you would like to view on the left-hand side to interact with ready-made tables. Alternatively, double click on the dataset and you will be able to build your bespoke table.
User Guidance
User guidance is available in the "help" section which can be found in the top right hand side with a question mark icon or at the bottom right hand side with the label "Need Help?".
The following video outlines some of the basic functionality available within Stat-Xplore online tool and shows users how to create and edit a simple table, as well as other basic features available within the tool such as downloading data in different formats, saving a user-made table and also how to present data using different chart options. Note that not all the functionality is available unless you are a registered user, such as saving tables.
Data Confidentiality
For the datasets available on Stat-Xplore, statistical disclosure control is applied which guards against the identification of an individual claimant. For futher information see Data Confidentiality.
The introduced random error method applied to the data from August 2024 will have a small impact on already published historic figures across Stat-Xplore releases. This update was necessary to enable a move to a more advanced data processing platform to maintain data confidentiality.
Moving to Census Output Area 2021 Geographies
Unless specified, most of the geographies available on the Stat-Xplore datasets currently use the 2011 Census Output Areas (COAs). Development work and testing is underway to move our statistical publications to report on 2021 COA geographies. This will be on a rolling basis starting for some publications from Autumn 2025. Users will be informed of updates via the DWP Statistical Work
Feedback/Enquiry
Need Help?
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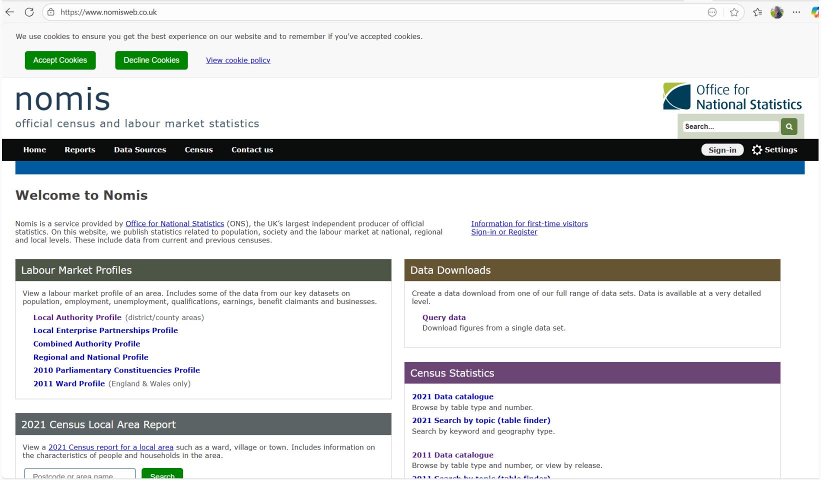
Accessibility Statement
Stat-Xplore - Powered by SuperSTAR

Nomis



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Information for first-time visitors
Sign-in or Register
Labour Market Profiles
View a labour market profile of an area. Includes some of the data from our key datasets on population, employment, unemployment, qualifications, earnings, benefit claimants and businesses.
Local Authority Profile (district/county areas)
Local Enterprise Partnerships Profile
Combined Authority Profile
Regional and National Profile
2010 Parliamentary Constituencies Profile
2011 Ward Profile (England & Wales only)
2021 Census Local Area Report
View a 2021 Census report for a local area such as a ward, village or town. Includes information on the characteristics of people and households in the area.
Postcode or area name
Search
Data Downloads
Create a data download from one of our full range of data sets. Data is available at a very detailed level.
Query data
Download figures from a single data set.
Census Statistics
2021 Data catalogue
Browse by table type and number.
2021 Search by topic (table finder)
Search by keyword and geography type.
2011 Data catalogue
Browse by table type and number, or view by release.

Tips on working with geographical data

- These data are usually open and under OGL Licence (mention it in your application if you want to link them to Controlled Access or other data)
- LADs codes changed between 2011 and 2021 → Lookup table on geoportal
- UK lower level area classification is slightly different from the one used in Europe (i.e. no direct correspondence with NUTS level geography)

Thank you for your attention!

Feel free to get in touch:

valentina.diiasio@soton.ac.uk